

Question	Answer	Marks
7(a)	Once the skateboarder is airborne, he is subjected to air resistance while the skateboard is mostly subjected to ground resistance. As these two resistive forces are not the same, the skateboarder and the skateboard will be subject to different deceleration. The longer the skateboarder is airborne (jumps higher), the further apart he and the skateboard would be upon landing and he may as a result miss the skateboard if this difference exceeds the length of the skateboard.	[1] [1] [1]
7(b)(i)	The missing force is exerted by the rear foot of the skater. This force is acting downwards where the left foot is. The magnitude of this force is such that the resultant force on the skateboard is zero.	[1] [1]
7(b)(ii)	In doing so, he exerts a slight downward force that generates a clockwise moment on the skateboard helping it to return to the horizontal position. As the surface of the deck is covered with grip tape, the frictional force along the surface of the deck helps to further lift the skateboard.	[1] [1]
7(c)(i)	From the graph, the power of the electric motor is 320 W when v is 4 m s^{-1}. Since $P = Fv$, the driving force provided by the electric motor is $F = 320/4 = 80 \text{ N}$ Since he is traveling at maximum speed, the resultant force must be zero, hence the frictional force that acts on him is also 80 N.	[1] [1] [1]
7(c)(ii)	From the graph, the driving force on the skateboarder when he is on surface A is given by $F = P/v = 540/6 = 90.0 \text{ N}$. The resistive force that acts on the skateboarder while he is on surface B can be found by using the dotted graph, where $F = P/v = 280/6 = 46.6 \text{ N}$ Hence when he moves onto the surface B, there is now a resultant force of $90.0 - 46.6 = 43.4 \text{ N}$ in the forward direction.	[1] [1] [1] [1]
7(c)(iii)	From the graph, the driving force on the skateboarder when he is on surface A is given by $F = P/v = 540/6 = 90.0 \text{ N}$. The resistive force that acts on the skateboarder while he is on surface B can be found by using the dotted graph, where $F = P/v = 280/6 = 46.6 \text{ N}$ Hence when he moves onto the surface B, there is now a resultant force of $90.0 - 46.6 = 43.4 \text{ N}$ in the forward direction.	[1] [1] [1]
7(c)(iv)	From Fig. 7.5, we can see that different combined mass can result in different slopes at the early part of the graph. Hence a large vertical force exerted by the skateboard on the ground can lead to larger ground resistance. Using Fig. 7.4, we can also say that the material of the surface can affect the ground resistance where surface A will result in <u>larger ground resistance</u> than surface B.	[1] [1]
7(d)	For a shorter truck, the deck will be nearer to the ground. Hence when the tail end is pressed onto the ground, the front end of the skateboard will be inclined at a smaller angle. For a short skateboarder, he will not need to bend his knees as much and hence find it easier to perform the Ollie.	[1] [1]
	Max Marks	21